

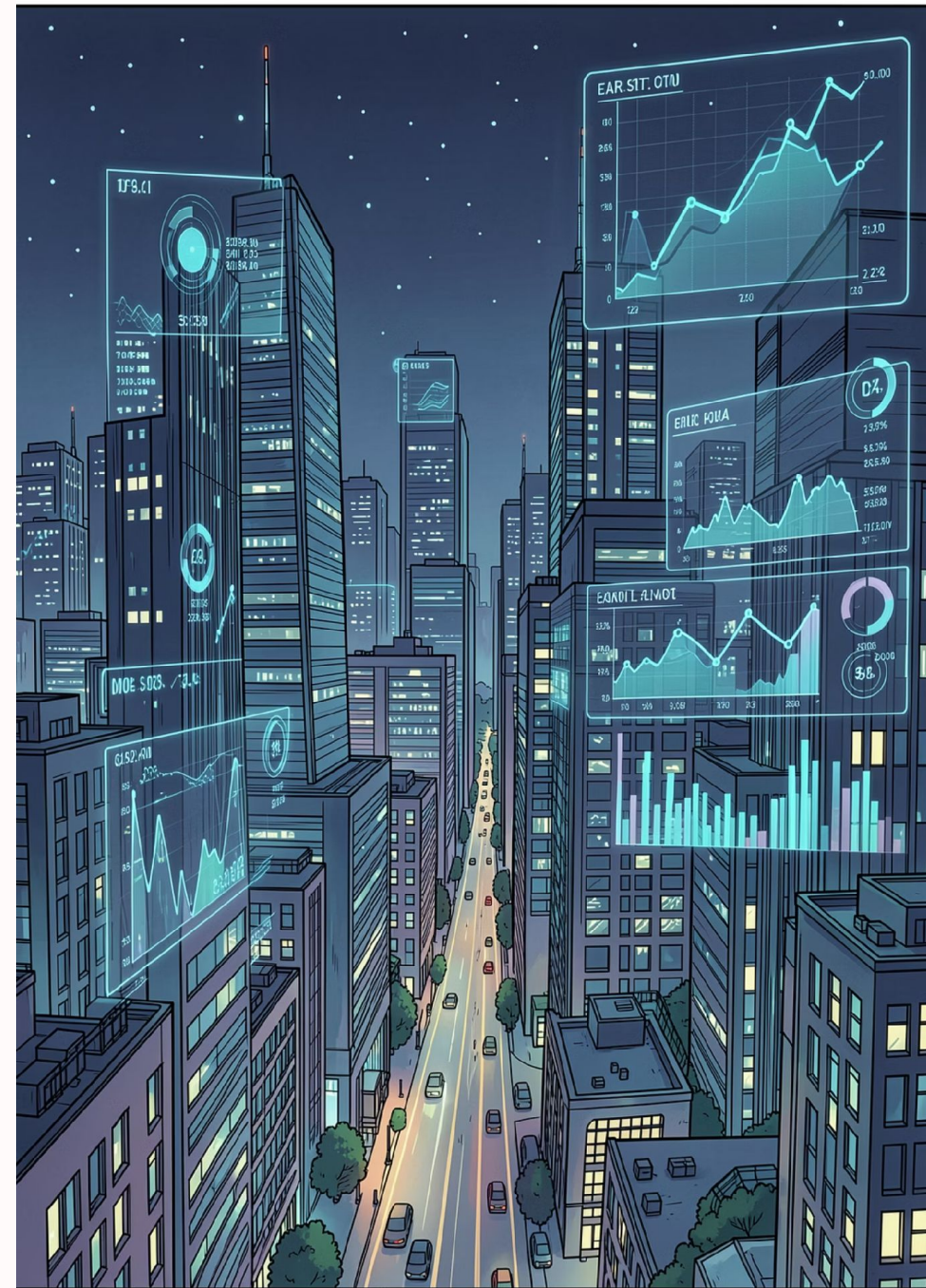
Global Airbnb Performance Dashboard

A 3-page interactive Power BI dashboard analyzing listing growth, market concentration, pricing dynamics, and host trust signals across 10 major global cities.

POWER BI PROJECT

BUSINESS INTELLIGENCE

10 CITIES





Project Overview

Airbnb's global dataset is rich — but raw and multi-dimensional. This project builds a **production-grade Power BI dashboard** transforming that data into a structured, navigable BI product with dynamic views, bookmark-based UI toggling, and conditional formatting.

Business Strategists

Track market concentration and growth lifecycle

Marketing Teams

Understand seasonal review patterns and city performance

Operations Teams

Analyze host trust, pricing tiers, and rating gaps

Dashboard Structure

The project is organized as a **3-page interactive dashboard**, each page solving a distinct analytical problem.

1

New Listings Growth

Historical lifecycle analysis from 2008–2021, annotated with growth phase markers

2

Market Share & Pricing

City concentration, Superhost vs. non-Superhost distribution, and average pricing by property type

3

Ratings & Review Behavior

City-level ratings, review frequency, seasonal patterns, and host trust analysis

Key Metrics at a Glance

279K

Total Listings

Across all cities and property types

10

Cities Analyzed

Global markets from Paris to Hong
Kong

182K

Total Hosts

Active hosts across the platform

144

Property Types

Unique accommodation categories

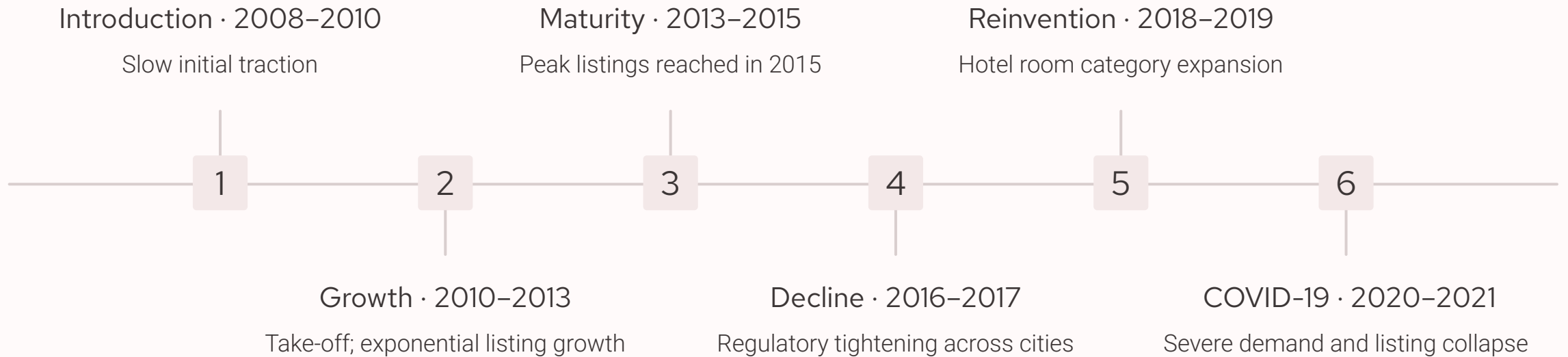
5.3M+

Total Reviews

Review events captured in the dataset

New Listings Growth Lifecycle (2008–2021)

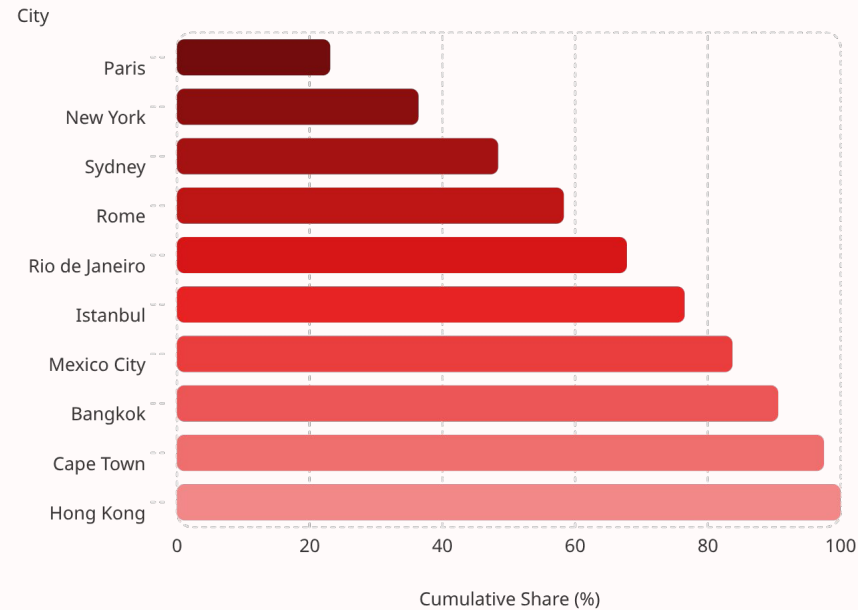
How has Airbnb's listing volume evolved over time, and what external forces shaped that growth?



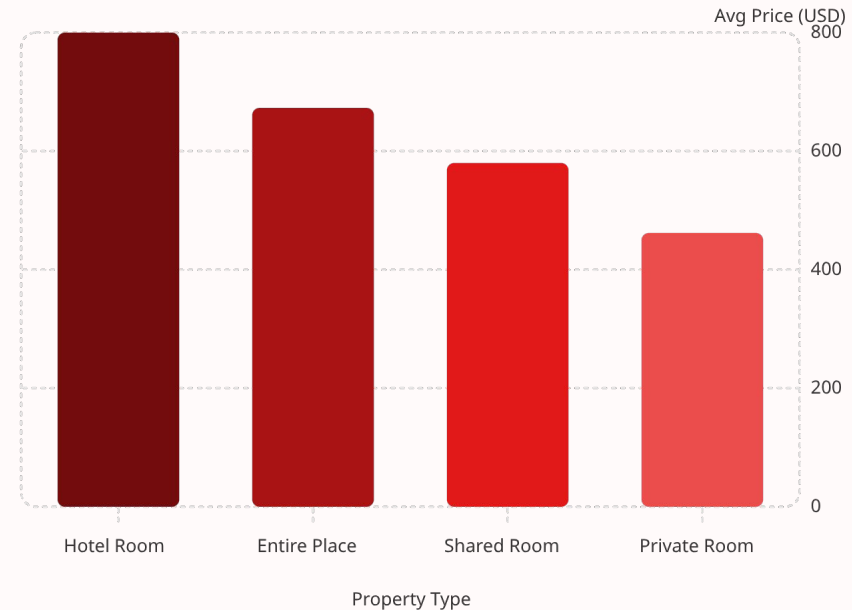
 **Key Insight:** 2015 saw the highest new listings globally, driven by Entire Place. Airbnb became profitable in H2 2016 — despite declining listings. COVID-19 collapsed all categories simultaneously.

Market Share by City & Pricing

Pareto Analysis – Cumulative Market Share



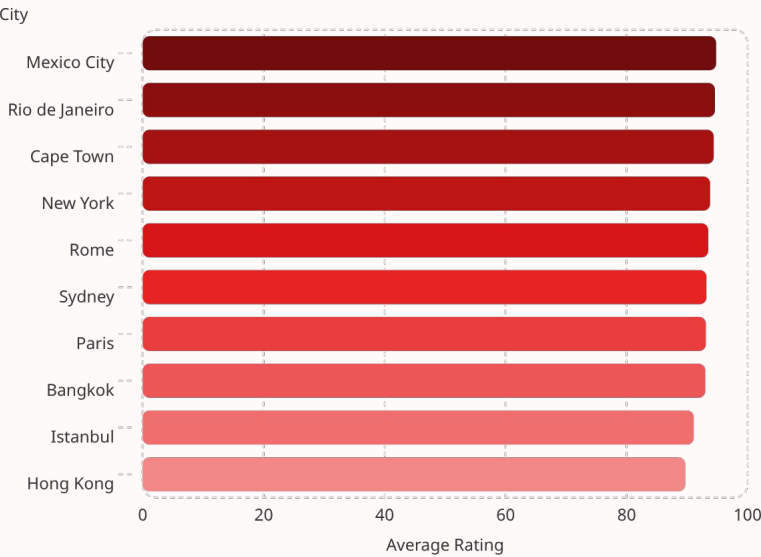
Average Price by Property Type



Key Insight: Paris, New York, and Sydney account for **48.4% of all listings**. Hotel rooms average **\$800/night** vs. Airbnb Entire Place at **\$673** — a 19% premium driving travelers toward Airbnb. Shared rooms show a pricing inefficiency: only a \$118 gap vs. private rooms despite significantly less privacy.

Ratings & Review Behavior

Overall Ratings by City



Host Trust Signal Matrix

	No Photo	Photo
Not Verified	0.1%	26.5%
Verified	0.1%	73.2%

Best & Worst Rated

Mexico City and Rio de Janeiro lead. Hong Kong and Istanbul lag. **Cleanliness** and **Value for Money** score lowest globally — the most actionable improvement areas.

Review Frequency

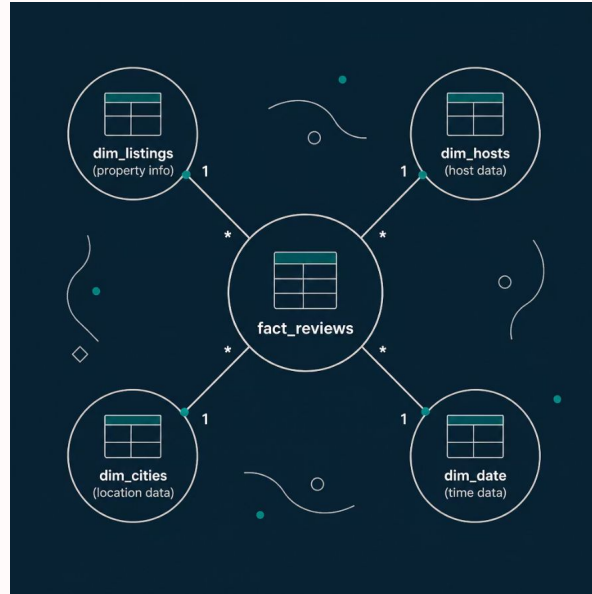
86.3% of reviewers wrote only once — review count is highly skewed. A 283-review outlier (Bangkok, Oct 2020) was flagged as a data anomaly. Filter threshold: reviews > 85.

Seasonal Patterns

Paris and Rome dominate **April–August**. New York surges **November–December**. Sydney shows consistent year-round activity with mild winter dips.

Technical Implementation

Star Schema Data Model



Data Cleaning Pipeline

Relationships validated for referential integrity. Star schema ensures fast DAX query performance and clean slicing across all visuals.

01

Missing Value Handling

Null imputation for price, rating, and host fields

03

Date Parsing

Converted string date columns to proper Date type

05

Outlier Flagging

Identified 283-review anomaly for manual review

02

Categorical Normalization

Standardized property_type and room_type labels

04

Review Count Filtering

Applied >85 threshold to remove noise listings

06

Trust Signal Encoding

Derived binary flags for identity_verified and has_photo

DAX Measures Engineered

Core DAX measures powering the dashboard's analytical depth.

Cumulative Listings

Running total of listings over time using **CALCULATE** and **FILTER(ALL(dim_date[year]))** for lifecycle phase tracking.

% Monthly Reviews by City

City's share of total monthly reviews via **DIVIDE** and **ALL(dim_cities[city])** for seasonal comparison.

Avg Daily Price by Property Type

Average price per category using **AVERAGEX** over the listings table for pricing benchmarking.

Fully Verified Hosts %

Share of hosts with verified identity *and* profile photo using **FILTER** with compound boolean conditions.

Strategic Takeaways

This dashboard translates raw Airbnb data into actionable intelligence across three analytical dimensions.

Market Concentration

Paris, New York, and Sydney drive **48.4%** of all listings. A 19% hotel price premium fuels Airbnb demand in Paris.

Trust & Quality

73.2% of hosts are fully verified. Cleanliness and Value for Money are the lowest-scoring dimensions globally — the highest-priority improvement levers.

Growth Lifecycle

Airbnb reached profitability in H2 2016 despite declining listings. COVID-19 collapsed all categories simultaneously, revealing systemic demand vulnerability.

